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Faculty of Engineering, Built Environment and
Information Technology

EAI 320

INTELLIGENT SYSTEMS

PRACTICAL 4

Name and Surname	Student Number	Signature	% contribution
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I. CODE

Listing 1: Naïve Bayes classifier.

```
1 % Intelligent Systems EAI 320 Practical 4 (NAive Bayes)
2 % Moosa Osman u24566722
3 % Last updated: 20260516
4
5 function rps = agent_bayes(previous , n_rounds)
6     persistent opponentHistory;
7     persistent ourHistory;
8
9     % General procedure:
10    % Random for move one ==> start predicting based on previous
        moves
11    % More moves ==> greater chance of predicting move properly
12    % perform Naive Bayes and find most likely move, then choose
        winning move
13
14    if isempty(previous)
15        opponentHistory = [];
16        ourHistory = [];           % init the stuff
17
18        rps = randi([0, 2]);       % play random first time
19        ourHistory = [ourHistory; rps];
20        return;
21    end
22
23    % update the history
24    opponentHistory = [opponentHistory; previous];
25    N = length(opponentHistory);
26
27    % extracting the features
28    f1Hist = opponentHistory(1:end-1);           % f1: Opponent's move
        1 round ago
29    f2Hist = ourHistory(1:end-1);                 % f2: Our move 1 round ago
30    cHist = opponentHistory(2:end);               % C: Opponent's move
        this round
31
32    currentf1 = opponentHistory(end);
33    currentf2 = ourHistory(end);
34
35    posteriorScores = zeros(3, 1);               % values for the
        posteriors, to be used later
36    totalSamples = length(cHist);                 % no of samples
37
38    for idx = 1:3                                % for all: need to find Cond. probs.
39        classVal = idx - 1;
40
41        countCk = sum(cHist == classVal);         % Count occurrences of
        the target class
42
43        prior = (countCk + 1) / (totalSamples + 3); % have to add
        +1 and +3 otherwise NaN error on clickUP grader
44        % this is caused by values being 0, so for countCk = 1,
        totalSamples = 0 ==> problem!, but now will evaluate to
        0.667
```

```

45
46     %find cond. prob. P(f1|Ck) i.e. last move
47     countf1Givenk = sum(f1Hist(cHist == classVal) == currentf1);
48     pf1Givenk = (countf1Givenk + 1) / (countCk + 3);
49
50     % P(f2 | Ck) i.e. one move ago
51     countf2Givenk = sum(f2Hist(cHist == classVal) == currentf2);
52     pf2Givenk = (countf2Givenk + 1) / (countCk + 3);
53
54     posteriorScores(idx) = prior * pf1Givenk * pf2Givenk;      %
        multiply for p(f|Ck)p(Ck)
55 end
56
57 [~, max_idx] = max(posteriorScores);      % for argmax part of Prob
58 predicted_opponent_move = max_idx - 1;
59
60 rps = mod(predicted_opponent_move + 1, 3);      % winning
        move
61
62 ourHistory = [ourHistory; rps];      % update for nxt
        iteration
63
64 % Results (10 rounds)
65 % Run          Time (s)      Rounds Won      Matches Won
        Anomilies
66 % 1            19.062877      648517      700      Only
        49871 rounds for no_repeat, but still 100 matches won
67 % 2            16.313442      648523      700      Only
        49854 rounds for no_repeat, but still 100 matches won
68 % 3            20.008488      648603      700      Only
        49917 rounds for no_repeat, but still 100 matches won
69 % 4            22.458893      648504      700      Only
        49870 rounds for no_repeat, but still 100 matches won
70 % 5            20.65606       648524      700      Only
        49891 rounds for no_repeat, but still 100 matches won
71 % 6            20.963052      648418      700      Only
        49750 rounds for no_repeat, but still 100 matches won
72 % 7            20.414442      648195      700      Only
        49499 rounds for no_repeat, but still 100 matches won
73 % 8            18.963757      648618      700      Only
        49985 rounds for no_repeat, but still 100 matches won
74 % 9            18.798465      648687      700      Only
        50025 rounds for no_repeat, but still 100 matches won
75 % 10           19.613149      648338      700      Only
        49681 rounds for no_repeat, but still 100 matches won
76 % Average      19.7252625      648492.7      700
77
78 % Results show that Naive Bayes performs relatively well for most
        agents
79 % For non deterministic agents (no_repeat) it doesn't perform the
        best but nevertheless all matches are still won
80 % For other agents, it can correctly predict the move that wioll
        be played and counter it, winning 92.641814286% of rounds
81 % Compared to prac 2, Naive Bayes has better results, but took
        longer to run and was easier to implement
82 % Compared to prac 3, NB has roughly the same results as the
        final optimised NN, but took twice as long to run. But it was

```

```
            much easier to code
83      % Benefit of Naive Bayes is evident: easy to implement with
            results that are relatively good
84
85  end
```